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Degree: B.Tech

Roll no.:.....
Semester: 3

Mid Semester Examination, September 2026

Course Code: ITMTC301/INMTC301/EIMTC301
Course title: Optimization Principles and Techniques

Time: 1 hour 30 minutes

Maximum Marks. 20

Note: Attempt all the questions in given order only. Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Solve the given LPP using the simplex method. Max Z = $-x + 2y$ subject to $x + y \leq 6$ $2x + y \leq 8$ $x, y \geq 0$	2.5	CO1
1b	A firm manufactures headache pills in two sizes A and B. Size A contains 2 grains of aspirin, 5 grains of bicarbonate, and 1 grain of codeine. Size B contains 1 grain of aspirin, 8 grains of bicarbonate and 6 grains of codeine. It is found by user that it requires at least 12 grains of aspirin, 74 grains of bicarbonate and 24 grains of codeine for providing immediate effect. It is required to determine the least number of pills a patient should take to get immediate relief. Formulate the problem as a standard LPP.	1.5	CO1
2a	Obtain all the basic feasible solutions of the following system of linear equations: $2x_1 + x_2 - x_3 = 2$ $3x_1 + 2x_2 + x_3 = 3$ Check if the system has degenerate solution or not.	2	CO1
2b	Find the minimum value of the LPP using graphical method. Min Z = $x_1 + 0.5x_2$ subject to $3x_1 + 2x_2 \leq 12$ $5x_1 \leq 10$ $x_1 + x_2 \geq 8$ $-x_1 + x_2 \geq 4$ $x_1, x_2 \geq 0$	2	CO1
3a	Use two-phase simplex method to solve the following LPP. Max Z = $3x_1 + 2x_2$ subject to $2x_1 + x_2 \leq 2$ $3x_1 + 4x_2 \leq 12$ $x_1, x_2 \geq 0$	2	CO2
3b	We have to assign three jobs, I, II, III to three workers A, B, and C. The time taken by different workers (in hours) in completing different jobs is given below: $\begin{array}{cccc} & I & II & III \\ A & 8 & 7 & 6 \\ B & 5 & 7 & 8 \\ C & 6 & 8 & 7 \end{array}$ Find the mimimum time (in hours) taken by the workers to complete all jobs.	2	CO2

4a	Max Z = $-x_1 + 2x_2$ subject to $3x_1 + 4x_2 \leq 12$ $2x_1 - x_2 \geq 2$ $x_1, x_2 \geq 0$ State and solve the dual problem graphically. Using the duality theorems obtain the optimal solution of primal problem from the optimal solution of the dual problem.	2	CO2
4b	Use dual simplex method to solve the following LPP. Min Z = $2x_1 + x_2$ subject to $3x_1 + x_2 \geq 3$ $4x_1 + 3x_2 \geq 6$ $x_1 \geq 0, x_2 \geq 0$	2	CO2
5a	Solve the given transportation problem using Vogel's Approximation Method (VAM). *D1* *D2* *D3* *D4* *Supply* *O1* 10 0 20 11 15 *O2* 1 7 9 20 25 *O3* 12 14 16 18 5 *Demand* 12 8 15 10	2	CO2
5b	Apply UV method to find the optimal solution of the transportation problem given in 5a.	2	CO2